

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester II)
Scala for Data Science Practical

Practical - 14/15/16/17

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Class :SYIT	Batch:01
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Practical 14

Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

INPUT:

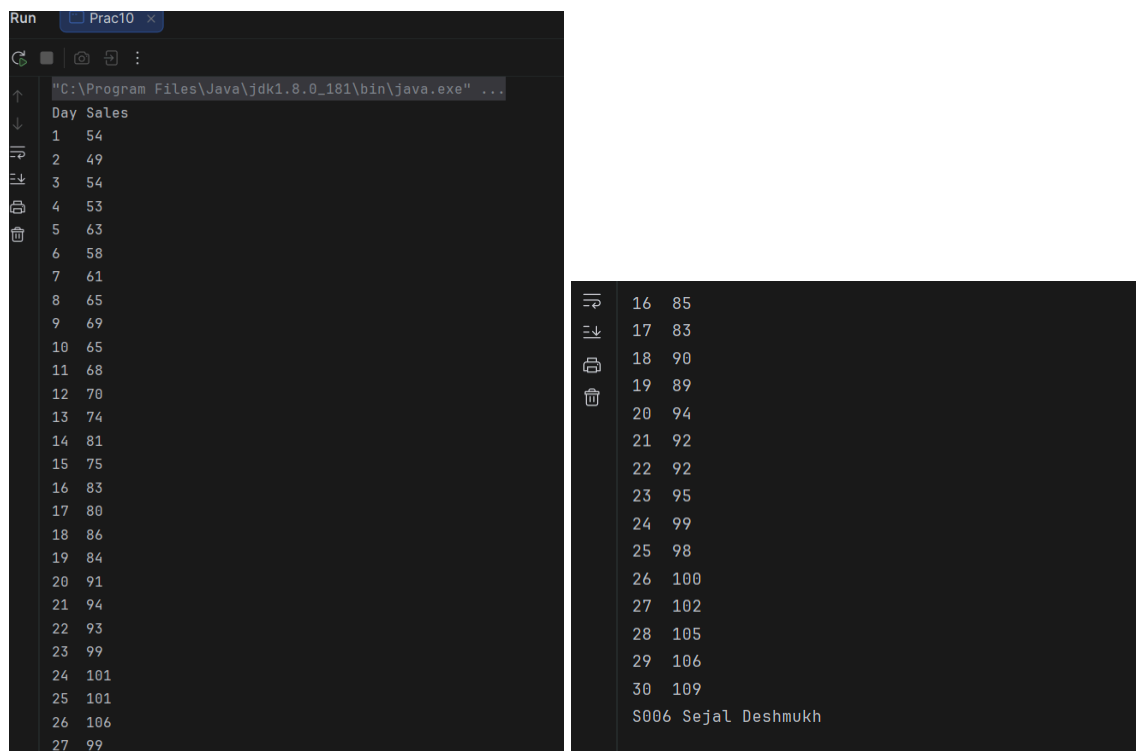
```
import scala.util.Random

object Prac10 extends App {

  println("Day\tSales")

  for (day <- 1 to 30) {
    val sales = 50 + day * 2 + Random.nextInt(10) - 5
    println(s"$day\t$sales")
  }
}
```

OUTPUT:



The screenshot shows a Java application window titled "Prac10" with a command prompt interface. The output consists of two parts: a table of Day vs Sales and a list of numbers.

Day	Sales
1	54
2	49
3	54
4	53
5	63
6	58
7	61
8	65
9	69
10	65
11	68
12	70
13	74
14	81
15	75
16	83
17	80
18	86
19	84
20	91
21	94
22	93
23	99
24	101
25	101
26	106
27	99

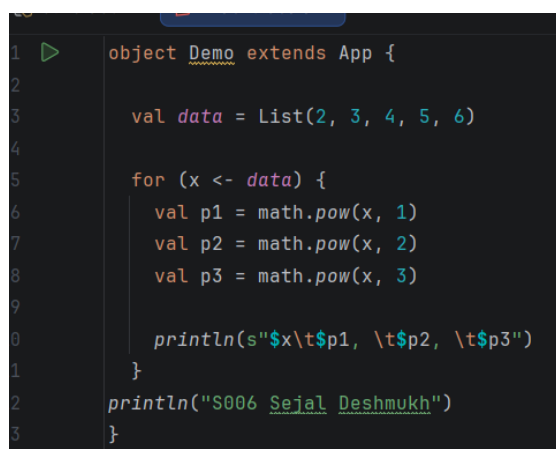
16	85
17	83
18	90
19	89
20	94
21	92
22	92
23	95
24	99
25	98
26	100
27	102
28	105
29	106
30	109

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Practical 15

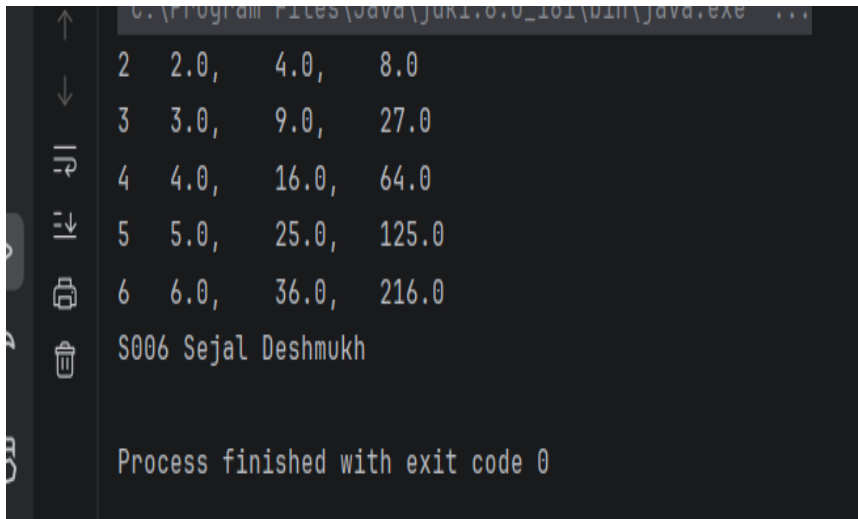
Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3.

INPUT:



```
1  object Demo extends App {  
2  
3    val data = List(2, 3, 4, 5, 6)  
4  
5    for (x <- data) {  
6      val p1 = math.pow(x, 1)  
7      val p2 = math.pow(x, 2)  
8      val p3 = math.pow(x, 3)  
9  
10     println(s"$x\t$p1, \t$p2, \t$p3")  
11   }  
12   println("S006 Sejal Deshmukh")  
13 }
```

OUTPUT:

A screenshot of a Java IDE window. The title bar shows the path 'C:\Program Files\Java\jdk1.8.0_101\bin\java.exe'. The main area displays the output of a program. On the left, there is a vertical toolbar with icons for undo, redo, and other editing functions. The output text is as follows:

```
2 2.0, 4.0, 8.0
3 3.0, 9.0, 27.0
4 4.0, 16.0, 64.0
5 5.0, 25.0, 125.0
6 6.0, 36.0, 216.0
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Process finished with exit code 0
```

Practical 16

16) Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.

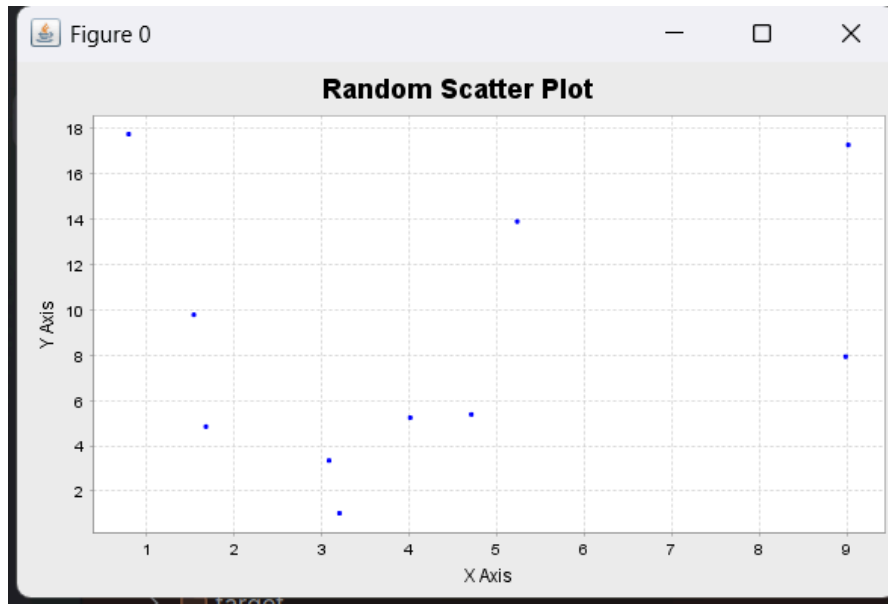
INPUT:

```

1  import breeze.plot._
2  import scala.util.Random
3
4  object Demo extends App {
5      val x = List.fill(10)(Random.nextDouble() * 10)
6      val y = List.fill(10)(Random.nextDouble() * 20)
7      println(x)
8      println(y)
9
10     val f = Figure()
11     val p = f.subplot(0)
12
13     p += plot(x, y, '.', colorcode = "blue")
14
15     p.xlabel = "X Axis"
16     p.ylabel = "Y Axis"
17     p.title = "Random Scatter Plot"
18
19     f.refresh()
20     println("S006 Sejal Deshmukh")
21 }

```

OUTPUT:



```

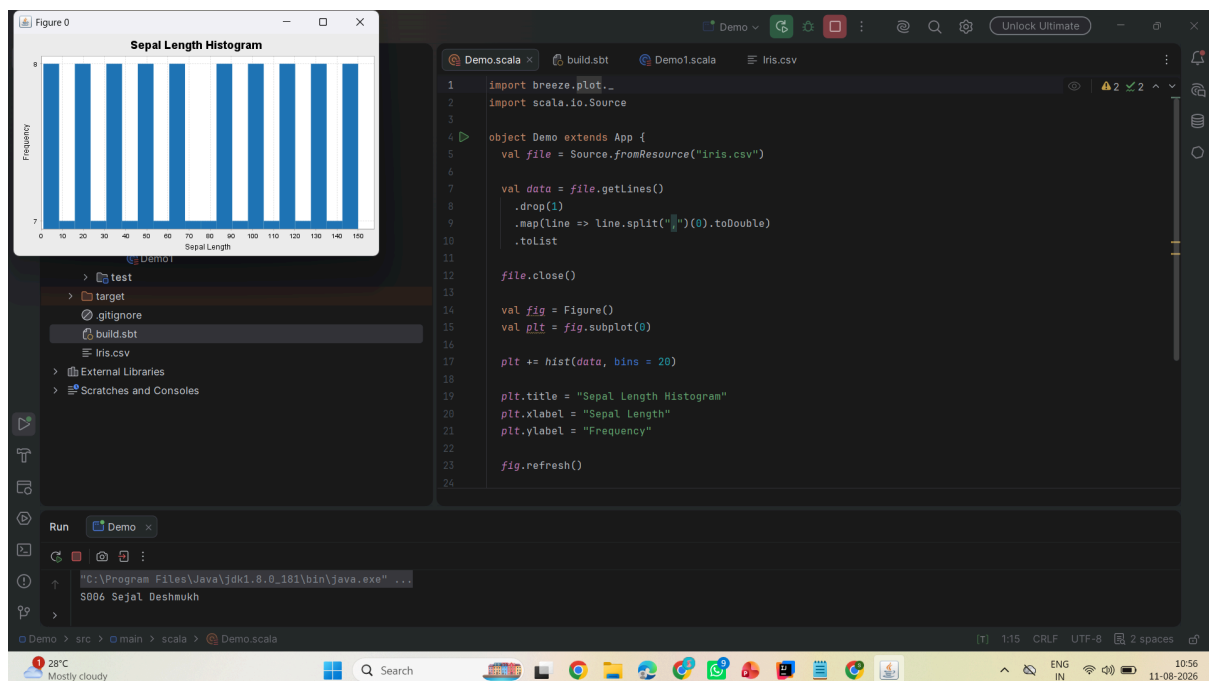
Run Demo x
[C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
List(4.709920702256838, 5.234002934008247, 3.2045514647824316, 3.0852524089922384, 9.014132119647314, 0.7975034809735604, 4.012201862189995, 1.541279019003312, 1.6812159889427,
List(5.381277360467651, 13.88698394764482, 1.0273173065575913, 3.349720500061766, 17.262323902220515, 17.736615457755516, 5.23841908920704, 9.771595018632524, 4.84172321804016,
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```

47314, 0.7975034809735604, 4.012201862189995, 1.541279019003312, 1.6812159889427492, 8.983661672892083)
20515, 17.736615457755516, 5.23841908920704, 9.771595018632524, 4.841723218040162, 7.930499664294173)

Practical 17

17) Generate a histogram of a dataset using Breeze-viz. Experiment with different bin sizes.



Practical 18

18) Plot a line graph for a dataset showing a trend over time.

